

$$\Leftrightarrow \textcircled{3} \quad (\because \theta = \frac{x}{2} = \frac{a}{2n}, \theta \in (0, \frac{\pi}{4n}])$$

3. prove 2.③:

$$s(2n+1)\theta - s\theta < \frac{s\theta}{2} s(2n)\theta < s(2n-1)\theta + s\theta$$

left: mid: right:

$$\theta \in (0, \frac{\pi}{4n}], n \geq 1$$

$$\text{proof: } 1) \quad s(2n+1)\theta = s(2n)\theta + c(2n)\theta s\theta < s(2n)\theta + s\theta$$

($\because c\theta < \frac{s\theta}{2}$)

$$\therefore s(2n+1)\theta - s\theta < \frac{s\theta}{2} s(2n)\theta$$

$$\therefore \text{left} < \text{mid}$$

$$2) \quad \because s(2n-1)\theta = s(2n)\theta - c(2n)\theta s\theta$$

$$\therefore s(2n-1)\theta + s\theta = s(2n)\theta(c\theta + \frac{s(1-c)\theta}{s(2n)\theta})$$

$$\therefore c\theta = c\theta - s\theta = -s^2\theta \therefore 1-c\theta = 2s^2\theta$$

$$\therefore \frac{1-c(2n)\theta}{s(2n)\theta} = \frac{s\theta}{c\theta}$$

$$\therefore \text{right} = s(2n)\theta(c\theta + \frac{s(1-c)\theta}{s(2n)\theta}) = s(2n)\theta \frac{c\theta + s\theta}{c\theta} = s(2n)\theta \frac{c(\theta - n\theta)}{c\theta} = s(2n)\theta \frac{c(n-1)\theta}{c\theta} = s(2n)\theta \frac{c(n-1)\theta}{c\theta}$$

$$\text{If } s^2\theta \neq 0, \therefore \sum_{k=1}^n c_k x_k = \frac{s(n+\frac{1}{2})x - s^2\theta}{2s^2\theta}$$

$$\therefore \sum_{k=1}^n c_k x_k = \frac{s(n+\frac{1}{2})x - s^2\theta}{2s^2\theta}, n \geq 2$$

$$\therefore \sum_{k=1}^n c_k x_k = \frac{s(n+\frac{1}{2})x + s^2\theta}{2s^2\theta}, n \geq 1$$

$$\therefore \text{if } x \neq 2m\pi, m \in \mathbb{Z}, \sum_{k=1}^n c_k x_k = \frac{s(n+\frac{1}{2})x - s^2\theta}{2s^2\theta}, n \geq 1$$

$$\text{and } \sum_{k=0}^n c_k x_k = \frac{s(n+\frac{1}{2})x + s^2\theta}{2s^2\theta}, n \geq 1$$

$$\textcircled{2} \Leftrightarrow x \sum_{k=1}^n c_k x_k < s\theta < x \sum_{k=0}^n c_k x_k, x = \frac{a}{n} \quad \textcircled{1} \int_0^a c(x) dx = s\theta, a \in (0, \frac{\pi}{2}]$$

$$\Leftrightarrow \frac{a}{n} \sum_{k=1}^n c_k \frac{a}{n} < s\theta < \frac{a}{n} \sum_{k=0}^n c_k \frac{a}{n} \quad \textcircled{3} \Leftrightarrow \frac{a}{n} \sum_{k=1}^n c_k \frac{a}{n} < s\theta < \frac{a}{n} \sum_{k=0}^n c_k \frac{a}{n}, n \geq 1, a \in (0, \frac{\pi}{2}]$$

$$\Leftrightarrow x \frac{s(n+\frac{1}{2})x - s^2\theta}{2s^2\theta} < s\theta < x \frac{s(n+\frac{1}{2})x + s^2\theta}{2s^2\theta} \quad \textcircled{3} \Leftrightarrow s(2n+1)\theta - s\theta < \frac{s\theta}{2} s(2n)\theta < s(2n-1)\theta + s\theta$$

$$x = \frac{a}{n}$$

proof: we want to prove $\textcircled{1} \Leftrightarrow \textcircled{2} \Leftrightarrow \textcircled{3}$, $n \geq 1$

If $\textcircled{1} \Rightarrow \textcircled{2}$

If $\textcircled{2} \Rightarrow \textcircled{1} \because c(x) \searrow \text{on } [0, a], a \in (0, \frac{\pi}{2}] \Rightarrow \textcircled{1} \Leftrightarrow \textcircled{2}$

$$\boxed{\int_a^b c(x) dx = s\theta |^b_a}$$

$$2s^2\theta \sum_{k=1}^n c_k x_k = s(n+\frac{1}{2})x - s^2\theta, n \geq 1$$

$$\text{proof: } \therefore s\theta - s\theta = 2s^2\theta \frac{a\theta}{2}$$

$$\therefore s(k+\frac{1}{2})x - s(k-\frac{1}{2})x = 2s^2\theta c_k x$$

take $k=1, 2, \dots, n$ and add these equations,

$$s(1+\frac{1}{2})x - s(1-\frac{1}{2})x = 2s^2\theta c_1 x$$

$$s(2+\frac{1}{2})x - s(2-\frac{1}{2})x = 2s^2\theta c_2 x$$

;

$$\therefore s(n+\frac{1}{2})x - s^2\theta = 2s^2\theta \sum_{k=1}^n c_k x$$

$$\theta = \frac{a}{2n} \in (0, \frac{\pi}{4n}]$$

$$4.5 \int_b^{b+2\pi n} c x dx$$

$$= \sum_{k=0}^{n-1} \int_{b+2k\pi}^{b+2(k+1)\pi} c x dx$$

$$= 0$$

$$5. \int_0^a c x dx = sa, \quad a \in [-\frac{\pi}{2}, \frac{3\pi}{2}]$$

proof: 1) if $a \in (0, \frac{\pi}{2}]$, $\int_0^a c x dx = sa$ (by 2)

2) if $a = 0$, holds.

3) if $a \in [-\frac{\pi}{2}, 0]$, then $-a \in [0, \frac{\pi}{2}]$

$$\therefore \int_0^a c x dx = - \int_0^{-a} c(-x) dx$$

$$= - \int_0^{-a} c x dx$$

$$= -s(-a) = sa$$

4) if $a \in [\frac{\pi}{2}, \frac{3\pi}{2}]$

$$\int_0^a c x dx = \int_0^{\frac{\pi}{2}} c x dx + \int_{\frac{\pi}{2}}^a c x dx$$

$$= \int_0^{\frac{\pi}{2}} + \int_{\frac{\pi}{2}}^{a-\pi} c(x+\pi) dx$$

$$= \int_0^{\frac{\pi}{2}} + \int_{\frac{\pi}{2}}^{a-\pi} (-x) dx$$

$$= \int_0^{\frac{\pi}{2}} - \int_0^{a-\pi} + \int_0^{-\frac{\pi}{2}} = \frac{\pi^2}{2} - s(a-\pi) + s(-\frac{\pi}{2}) = sa$$

$$4.2 \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} c x dx$$

$$= \int_{-\frac{\pi}{2}}^0 c x + \int_0^{\frac{3\pi}{2}} c x$$

$$= s(-\frac{\pi}{2}) + s\frac{3\pi}{2} \quad (\because \int_0^a c x = sa, a \in [-\frac{\pi}{2}, \frac{3\pi}{2}])$$

$$= 0$$

$$4.3 \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} s x dx$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} s x$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} s x + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} s(x+\pi)$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} s x - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} s x$$

$$= 0$$

$$4.4 \int_{b-\frac{\pi}{2}}^{b+\frac{3\pi}{2}} c x dx, \quad \forall b \in \mathbb{R}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} c(x+b) dx$$

$$= \int (cx + sb) dx$$

$$= c \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} x - sb \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} s x$$

$$= 0$$

$$\therefore c n \theta = c(n-1) \theta c \theta - s(n-1) \theta s \theta$$

$$< c(n-1) \theta c \theta \quad (\because (n-1) \theta \in (0, \frac{\pi}{4n-4}])$$

$$\therefore s(n-1) \theta > 0$$

$$< c(n-1) \theta \frac{\theta}{s \theta} \quad (\because \theta \in (0, \frac{\pi}{4n-4}) \subset (0, \frac{\pi}{2}))$$

$$\therefore c \theta < \frac{s \theta}{s \theta}$$

$$\therefore \frac{s \theta}{\theta} < \frac{c(n-1) \theta}{c n \theta}$$

$$\Leftrightarrow \frac{s \theta}{\theta} s 2 n \theta < s 2 n \theta \frac{c(n-1) \theta}{c n \theta} \quad (\because s 2 n \theta > 0)$$

$$\Leftrightarrow \text{mid} < \text{right}$$

III

$$4.1 \int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} c x$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} c x + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} c x dx$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} c x + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} c(x+\pi)$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} c x - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} c x$$

$$= 0$$

2) $\forall a \in \mathbb{R}$,

$$\int_0^a x dx = \int_0^{\frac{\pi}{2}} x + \int_{\frac{\pi}{2}}^a x$$

$$= \int_0^{\frac{\pi}{2}} x + \int_0^{a-\frac{\pi}{2}} s(x+\frac{\pi}{2})$$

$$= \int_0^{\frac{\pi}{2}} x + \int_0^{a-\frac{\pi}{2}} cx$$

$$= 1 + s(a-\frac{\pi}{2})$$

$$= 1 - s(\frac{\pi}{2} - a)$$

$$= 1 - ca \quad \square$$

$$8. \int_a^b x dx = sb - sa,$$

$$\int_a^b \sin x dx = -(cb - ca)$$

$$6. \int_0^a cx dx = sa, \quad \forall a \in \mathbb{R}$$

$$\text{proof: } \forall a \in [-\frac{\pi}{2}, \frac{3\pi}{2}]$$

$$\exists n, s, t, \quad b = a - 2\pi n \in [-\frac{\pi}{2}, \frac{3\pi}{2}]$$

$$\therefore \int_0^a cx dx = \int_0^b cx dx + \int_b^a cx dx$$

$$= \int_0^b cx dx \quad (\text{by 4.5})$$

$$= sb$$

$$= s(a - 2\pi n)$$

$$= sa$$

$$7. \int_0^a x dx = 1 - ca, \quad \forall a \in \mathbb{R}$$

proof: 1) if $a = \frac{\pi}{2}$,

$$\int_0^a x = \int_0^{\frac{\pi}{2}} x = \int_{-\frac{\pi}{2}}^0 s(x + \frac{\pi}{2}) dx$$

$$= \int_{-\frac{\pi}{2}}^0 cx dx$$

$$= -s(-\frac{\pi}{2})$$

$$= 1 = 1 - ca$$